

BIOS 755: Parametric mean curves

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Parametric Curves

- ▶ An alternative approach for analyzing the parallel-groups repeated measures design is to consider parametric curves for the time trends.
- ▶ We model the means as an explicit function of time
 - ▶ linear terms: $time$,
 - ▶ polynomial terms: $time$, $time^2$,
 - ▶ log/square-root transform: $\log(time)$, $\sqrt{time}$,
 - ▶ restricted cubic splines or piecewise linear terms.

Linear Trend

- ▶ If the means tend to change linearly over time, we can fit the following model:

$$E(Y_{ij}) = \beta_0 + \beta_1 t_j + \beta_2 TRT_i + \beta_3 t_j TRT_i$$

- ▶ Let $Trt=1$ if the subject is in group 1 and 0 otherwise. For subjects in treatment group 1,

$$E(Y_{ij}) = \beta_0 + (\beta_1 + \beta_3)t_j + \beta_2$$

- ▶ For subjects in the control group 0

$$E(Y_{ij}) = \beta_0 + \beta_1 t_j$$

- ▶ Thus, each group's mean is assumed to change linearly over time.

Quadratic Trend

- ▶ If the means tend to change over time in a quadratic manner, we can fit the following model:

$$E(Y_{ij}) = \beta_0 + \beta_1 t_j + \beta_2 t_j^2 + \beta_3 TRT_i + \beta_4 t_j TRT_i + \beta_5 t_j^2 TRT_i$$

- ▶ For subjects in treatment group 1,

$$E(Y_{ij}) = \beta_0 + (\beta_1 + \beta_4)t_j + (\beta_2 + \beta_5)t_j^2 + \beta_3$$

- ▶ For subjects in the control group 0

$$E(Y_{ij}) = \beta_0 + \beta_1 t_j + \beta_2 t_j^2$$

- ▶ Thus, each group's mean is assumed to change quadratically over time.

Quadratic Trend

- ▶ To avoid problems of collinearity in the quadratic (or in any higher-order polynomial) trend model, one should always “center” t_j on its mean prior to the analysis, i.e., use

$$t_j^* = t_j - \bar{t} \quad \text{and} \quad t_j^{*2} = (t_j - \bar{t})^2$$

- ▶ For example, suppose $t_j = (1, 2, \dots, 10)$.
- ▶ Then the correlation between t_j and t_j^2 is 0.975 (no bueno).
- ▶ However, the correlation between t_j^* and t_j^{*2} is 0.

Log Trend

- ▶ If the means are monotonically increasing (or decreasing), but there appears to be “diminishing returns” in the changes, then a log transformation can be used

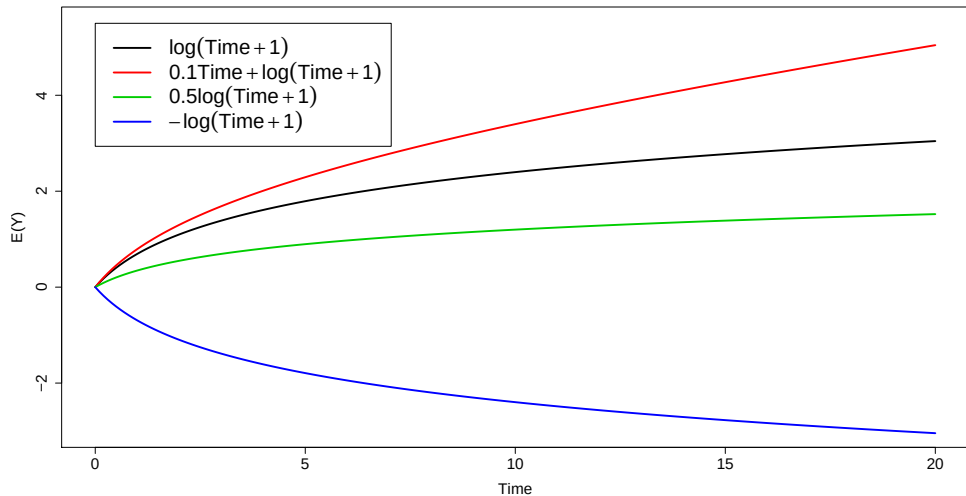
$$E(Y_{ij}) = \beta_0 + \beta_1 \log(t_j) + \beta_2 TRT_i + \beta_3 \log(t_j) TRT_i$$

- ▶ For subjects in treatment group 1,

$$E(Y_{ij}) = \beta_0 + (\beta_1 + \beta_3) \log(t_j) + \beta_2$$

- ▶ For subjects in the control group 0

$$E(Y_{ij}) = \beta_0 + \beta_1 \log(t_j)$$



Interpreting Coefficients with log-Transformations

- ▶ Model with **log-transformed predictor**

$$Y = \beta_0 + \beta_1 \log_e(X) + \varepsilon$$

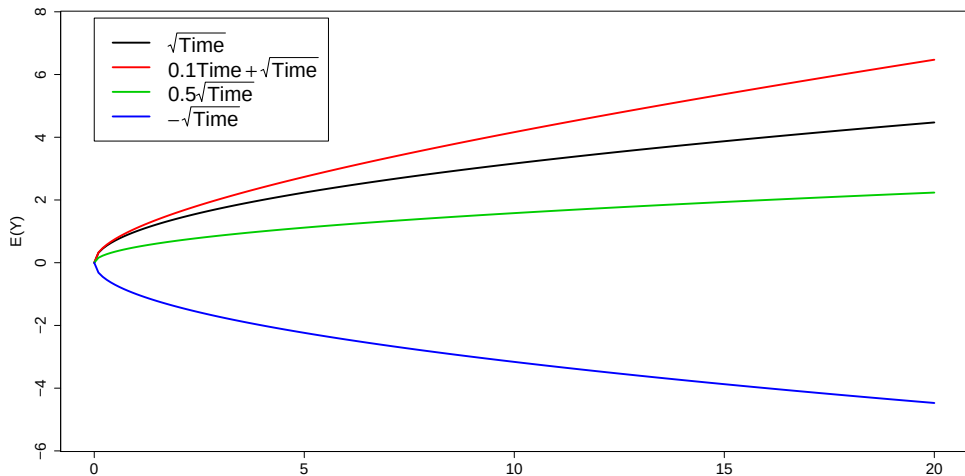
- ▶ **Percent-change shortcut:** a 10% increase in X (i.e. multiply by 1.1) changes Y by roughly $0.1 \beta_1$.
- ▶ Model with **log-transformed outcome**

$$\log_e(Y) = \alpha_0 + \alpha_1 X + \varepsilon$$

- ▶ Exponentiate to recover the multiplicative effect: e^{α_1} = ratio of expected Y 's.
- ▶ α_1 close to 0? Approximate percent change: $100(e^{\alpha_1} - 1)\%$ increase in Y per one-unit increase in X .

Square-root transformation

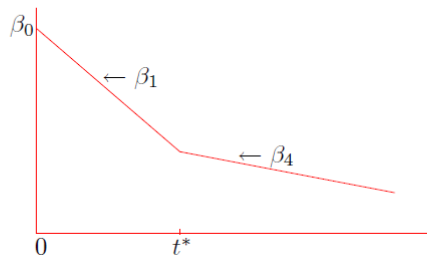
- The square root function $\sqrt{t}$ is another option for “diminishing returns”.



Linear Spline

- If the means change over time in a piecewise linear manner, we can fit the following linear spline model with knot at t^* :

$$\begin{aligned} E(Y_{ij}) &= \beta_0 + \beta_1 t_j + \beta_2 TRT_i + \beta_3 t_j TRT_i \quad t_j \leq t^* \\ &= \beta_0 + \beta_1 t^* + \beta_2 TRT_i + \beta_3 t^* TRT_i + \beta_4(t_j - t^*) + \beta_5(t_j - t^*) TRT_i \quad t_j > t^* \end{aligned}$$



Linear Spline

- For subjects in group 1

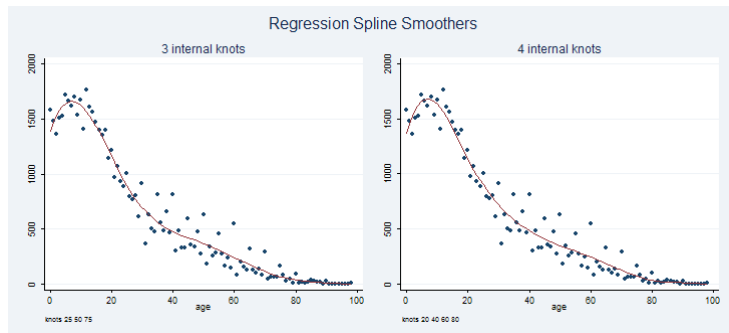
$$\begin{aligned} E(Y_{ij}) &= \beta_0 + \beta_2 + (\beta_1 + \beta_3)t_j \quad t_j \leq t^* \\ &= \beta_0 + \beta_2 + (\beta_1 + \beta_3)t^* + (\beta_4 + \beta_5)(t_j - t^*) \quad t_j > t^* \end{aligned}$$

- For subjects in group 2

$$\begin{aligned} E(Y_{ij}) &= \beta_0 + \beta_1 t_j \quad t_j \leq t^* \\ &= \beta_0 + \beta_1 t^* + \beta_4(t_j - t^*) \quad t_j > t^* \end{aligned}$$

Non-linear Spline

- ▶ Non-linear splines (e.g., cubic B-splines can also be used).



- ▶ These are used more with unbalanced data; we'll explore these with mixed models.

Summary of Features of Parametric Curve Models

- ▶ Allows one to model time trend and treatment effect(s) as a function of a few parameters.
- ▶ The treatment effect can be captured in one or two parameters, leading to more powerful tests when these models fit the data.
- ▶ Once you go beyond linear, quadratic, log, or square-root methods, the interpretations of the parameters are challenging (they are not easy for some of these!).
- ▶ Using profile analysis is usually preferred once things get beyond simple non-linear functions, especially for balanced data.

Summary of Features of Parametric Curve Models

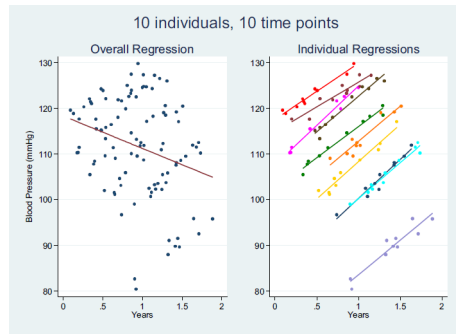
- ▶ For unbalanced data, parametric curve modeling is more common. As a result, we'll return to some of these when we get to mixed models.
- ▶ These models can tell the difference, for example, between a kid who is 6.0 and 6.9 years old.

Recall: Two Key Questions

- ▶ In longitudinal linear models, our first job is to specify the **mean response trajectory**.
- ▶ Doing so boils down to *two* fundamental questions:
 1. **Which** covariates should appear in the model?
 2. **How** should each chosen covariate be incorporated?

Why Do Other Variables Matter in Longitudinal Studies?

- ▶ **Goal:** estimate the exposure's effect on outcome trajectories without bias.
- ▶ Covariates related to both exposure *and* outcome (called confounders) can distort (confound) the estimate.
- ▶ Left unaddressed, confounding can:
 - ▶ Flip a slope's sign,
 - ▶ Shrink a true effect, or
 - ▶ Invent one where none exists.



Quick Confounder Rule of Thumb

1. **Is the covariate associated with the exposure?**
Check at baseline—and, if time-varying, at each wave.
2. **Is the covariate plausibly associated with the outcome?**
Use prior knowledge, literature, or simple bivariate models.
3. If **yes** to both, treat it as a candidate confounder and test its impact on the exposure coefficient.

Ignore variables unrelated to the exposure—they cannot confound it (though they may improve precision). Note: this implies we don't need to adjust for other variables when we randomize.

The “Change-in-Estimate” Test

- ▶ Fit the model **without** the covariate → obtain β_1 .
- ▶ Fit the model **with** the covariate → obtain β_1^* .
- ▶ **Decision rule:**
 - ▶ If $|\beta_1^* - \beta_1| / |\beta_1| \geq 10\%$ (or exceeds the SE of β_1), keep the covariate.
 - ▶ Otherwise, drop it—complexity without bias-reduction is unnecessary.

Mechanics in Linear Mixed Models

- ▶ **Baseline confounders:** include as fixed effects; they adjust all time points.
- ▶ **Time-varying confounders:**
 - ▶ Include as time-updated covariates only if *not* on the causal pathway, e.g., if

$$Exposure_{t-1} \rightarrow Covariate_t \rightarrow Outcome_{t+1}.$$

Then, adjusting for the covariate *reduces* the total exposure effect you want to estimate.

- ▶ Here, the covariate *mediates* the relationship between the exposure and outcome (mediators are not only time-varying).
- ▶ If the covariate is affected by prior exposure, consider marginal structural models (beyond the scope of this class).

Common Adjustment Pitfalls

Pitfall	Why It Hurts	Quick Fix
Over-adjustment (mediators)	Attenuates the exposure effect.	Use DAGs to distinguish mediators from confounders.
Colliders	Conditioning opens a spurious path, creating bias.	Avoid variables linked only through a common effect.
Precision $\neq$ Confounders	Improve SEs but don't change β_1 .	Include only if extra precision is needed & you have power to spare.

Practical Workflow Checklist

1. Plot & test exposure–covariate associations.
2. Fit base model; record β_1 .
3. Add candidate covariate; record β_1^* .
4. Compute change: $|\beta_1^* - \beta_1|/|\beta_1|$.
5. If $\geq 10\%$ (or >1 SE) $\Rightarrow$ keep covariate.
6. Document decisions for transparency & reproducibility.

Take-Home Messages

- ▶ Screen by exposure association first; if no link, no confounding.
- ▶ *Change-in-estimate* is a practical litmus test—focus on how the exposure coefficient moves.
- ▶ Use causal diagrams to avoid over-adjustment and collider bias.
- ▶ Thoughtful confounder control turns a good longitudinal analysis into a credible one.